THE UNITED REPUBLIC OF TANZANIA

MINISTRY OF EDUCATION, SCIENCE AND TECHNOLOGY

MENDGOAL SANGANA GIRL'S SEC SCHOOL

MINDTERM EXAMINATION

FORM THREE CHEMISTRY

TIME: 2;30 HOURS

INSTRUCTIONS:

(i). This paper consists of sections A, B, and C.

(ii). Answer all questions in sections A and B and only one question from section C.

(iii). Show all your work clearly.

(iv). Write your answers in the provided spaces.

SECTION A: MULTIPLE CHOICE QUESTIONS (10 Marks)

1. Choose the correct answer and write its letter in the answer booklet.

(i) A sample of water was found to form scum with soap but did not change upon boiling. This indicates that the water contains:

A. Calcium bicarbonate B. Magnesium chloride C. Sodium carbonate D. Hydrogen ions

(ii) A student added excess calcium carbonate to a dilute acid and observed effervescence. The gas produced was confirmed by: A. Turning limewater milky B. Relighting a glowing splint C. Changing blue litmus paper to red D. Forming a blue precipitate

(iii) The relative atomic mass of chlorine is 35.5. This is because: A. Chlorine has fractional atomic mass B. Chlorine exists as isotopes in nature C. Chlorine reacts with sodium to form salt D. Chlorine dissolves in water to form an acid

(iv) Which of the following statements is incorrect about acids and bases? A. Strong acids completely ionize in water B. Weak bases partially ionize in water C. Acids have a pH above 7 D. Bases can neutralize acids to form salts

(v) The oxidation state of chromium in $K_2Cr_2O_7$ is: A. +6 B. +3 C. +2 D. +7

SECTION B: STRUCTURED QUESTIONS (50 Marks)

3 (a) Define the following terms: (4 marks) i) Amphoteric oxides ii) Redox reaction iii) Molarity iv) Equivalent mass

(b) Balance the following chemical equations: (6 marks)

i) $MnO_4^- + Fe^{2+} + H^+ \rightarrow Mn^{2+} + Fe^{3+} + H_2O$

ii) $C_4H_{10} + O_2 \rightarrow CO_2 + H_2O$

iii) $NH_3 + O_2 \rightarrow NO + H_2O$

4(a) Explain why temporary hard water can be removed by boiling, while permanent hard water cannot. (4 marks)

(b) Mention four advantages and three disadvantages of water hardness (4 marks)

(c) Write an ionic equation to show the removal of temporary hardness by heating. (2 marks)

5 (a)

i) state dilution law

ii) What is the volume nitric acid that is 68% pure and has density of 1.42g/cm3 is needed to orepare 600cm3 of 0.4M of nitric acid

(b) 20Cm3 of 0.1M potassium hydroxide was titrated against 25cm3 of sulphuric acid.

Calculate concetration of sulphuric acid in

i)mol/dm3

ii) g/dm3

6 (a) Calculate the number of moles in 15.9 g of potassium permanganate ($KMnO_4$). (4 marks)

(b) A student reacted 10 g of calcium carbonate with dilute hydrochloric acid. Calculate the volume of carbon dioxide gas produced at room temperature and pressure (Molar volume = 24 dm^3, $CaCO_3$ = 100 g/mol). (6 marks)

SECTION C: ESSAY QUESTIONS (Choose One) (20 Marks)

7 (a) Explain the trend in atomic size across Period 3 of the periodic table. (6 marks)

(b) Discuss how ionization energy and electronegativity change across a period and down a group. (6 marks)

(c) Why are noble gases generally unreactive? Explain using electron configuration. (8 marks)

8. By using six examples, explain how the process of neutralization is important in day to day life. (chemical equations will be considered)